

Topic 13B: Entropy

12. understand that, since some endothermic reactions can occur at room temperature, enthalpy changes alone do not control whether reactions occur

13. know that entropy is a measure of the disorder of a system and that the natural direction of change is increasing total entropy (positive entropy change)

14. understand why entropy changes occur during:

- i changes of state
- ii dissolving of a solid ionic lattice
- iii reactions in which there is a change in the number of moles from reactants to products

Students should be able to discuss typical reactions in terms of disorder and enthalpy change, including:

- *dissolving ammonium nitrate crystals in water*
- *reacting ethanoic acid with ammonium carbonate*
- *burning magnesium ribbon in air*
- *mixing solid barium hydroxide, $\text{Ba(OH)}_2 \cdot 8\text{H}_2\text{O}$, with solid ammonium chloride.*

15. understand that the total entropy change in any reaction is the entropy change in the system added to the entropy change in the surroundings, shown by the expression:

$$\Delta S_{\text{total}} = \Delta S_{\text{system}} + \Delta S_{\text{surroundings}}$$

16. be able to calculate the entropy change for the system, ΔS_{system} , in a reaction, given the entropies of the reactants and products

17. be able to calculate the entropy change in the surroundings, and hence ΔS_{total} , using the expression:

$$\Delta S_{\text{surroundings}} = -\frac{\Delta H}{T}$$

18. know that the balance between the entropy change and the enthalpy change determines the feasibility of a reaction and is represented by the equation

$$\Delta G = \Delta H - T\Delta S_{\text{system}}$$

19. be able to use the equation $\Delta G = \Delta H - T\Delta S_{\text{system}}$ to:

- i predict whether a reaction is feasible
- ii determine the temperature at which a reaction is feasible

20. be able to use the equation $\Delta G = -RT \ln K$ to show that reactions which are feasible in terms of ΔG have large values for the equilibrium constant and vice versa

21. understand why a reaction for which the ΔG value is negative may not occur in practice